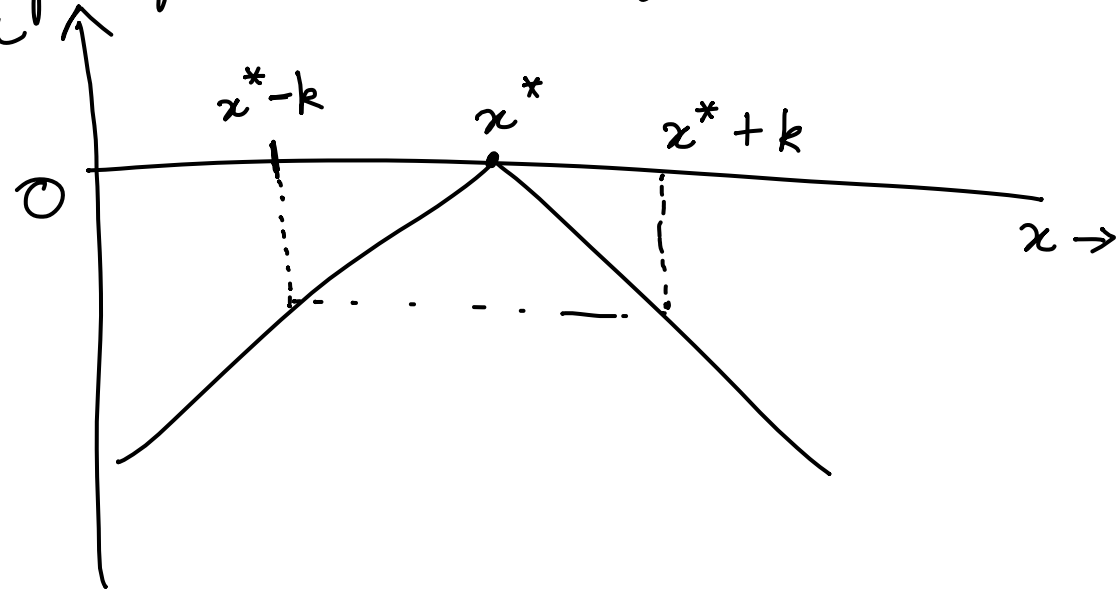


Hotelling Model of Electoral Competition

Voter's preferences are single-peaked :-

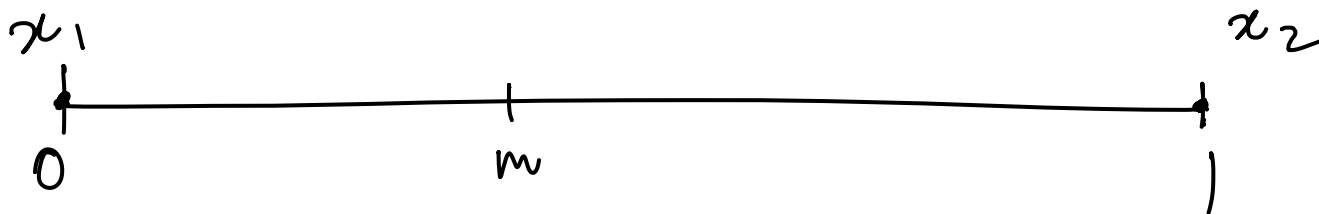


Electoral Game

Players :- The candidates (2 in no.)

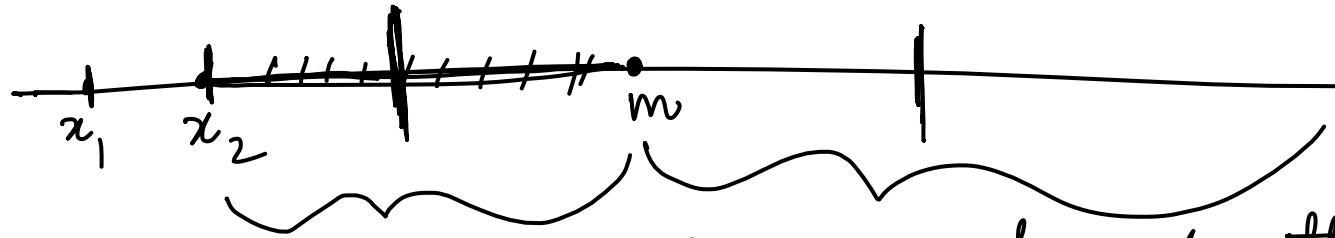
Action :- Each candidate's position

Payoffs :- If a candidate wins, she gets 2
 " " " is tied, " " 1
 " " " loses, " " 0



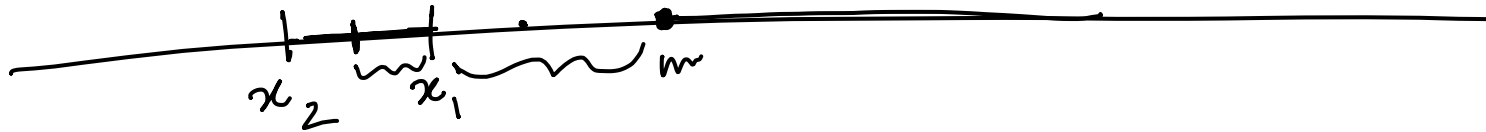
Fix x_2 .

Q:- Where should x_1 locate?

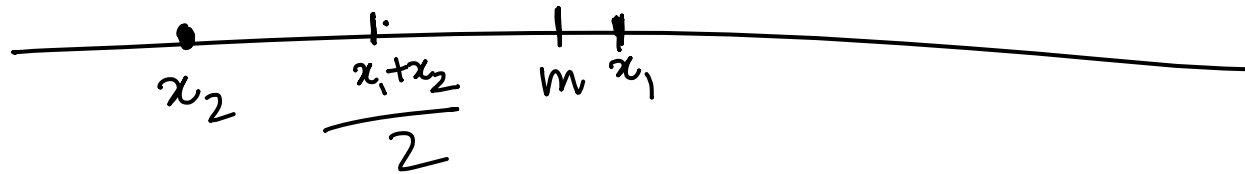


→ x_1 loses by locating anywhere to the left of x_2

What if x_1 locates b/w x_2 and m ?



Let x_2 remain fixed. Does 1 necessarily lose by locating right of the median?



Given x_2 ,
to win, x_1
must satisfy:-

$$\frac{x_1 + x_2}{2} < m$$

$$\Rightarrow x_1 < 2m - x_2$$

$$B_1(x_2) = \begin{cases} \{x_1 : x_2 < x_1 < 2m - x_2\} & \text{if } x_2 < m \\ \{m\} & \text{if } x_2 = m \\ \{x_1 : 2m - x_2 < x_1 < x_2\} & \text{if } x_2 > m \end{cases}$$

